

YI HUANG

Assistant Professor
Min H. Kao Department of EECS
Tickle College of Engineering
University of Tennessee, Knoxville

Email: yihuang@utk.edu
Office: (865) 974-3572
Address: 1520 Middle Drive, Room 431
Knoxville, TN 37996

Research Interests

- Hardware-software co-design for energy-efficient AI
- Neuromorphic computing based on emerging memory devices
- Intelligent sensing and communication systems for edge applications

Professional Experience

University of Tennessee, Knoxville

August 2026 – present

Assistant Professor

University of Massachusetts Amherst

September 2025 – July 2026

Postdoctoral Research Associate

Education

University of Massachusetts Amherst

September 2025

Ph.D. in Electrical and Computer Engineering

Advisor: Prof. Qiangfei Xia

Huazhong University of Science and Technology

June 2019 / June 2016

M.S. in Control Science and Engineering / B.E. in System Engineering

Advisor: Prof. Zhigang Zeng

Honors and Awards

- Ting-wei Tang Dissertation Prize of University of Massachusetts Amherst (2026)
- ECE David H. Navon Scholarship of University of Massachusetts Amherst (2022)
- ECE Graduate Fellowship of University of Massachusetts Amherst (2020)

Publications

Journal Articles (* indicate equal contribution)

1. Analog neural network-based wireless receiver
C. He*, **Y. Huang***, A. Korkmaz, A. Jaber Rad, Q. Xia, and L. Katehi
Science Advances, **12**, eaeg7153 (2026)
2. Single-step analog in-memory matrix computation in three-dimensional circuits
A. Jaber Rad, K. Kumar, F. Kiani, **Y. Huang**, A. Abdel-Maksoud, Z. Xiong, Y. Ling, W. Zhao, and Q. Xia
Advanced Intelligent Systems, e70500 (2026)
3. Hyperdimensional in-memory computing with analogue memristive crossbar arrays
Y. Huang*, A. J. Rad*, D. Belkin, N. Ge, J. Joshua Yang, M. Hu, and Q. Xia
Nature Communications, **17**, 9162 (2026)
4. Electrochromic hyperspectral embedding for ultracompact intelligent vision
R. Li, C. He, **Y. Huang**, E. Zhai, V. K. Sangwan, J. Zou, K. St. Luce, J. Cui, R. Kim, R. Liu, Z. Wang, X. Ling, H. Xie, X. Zhang, M. C. Hersam, Q. Xia, L. Katehi, and Y. C. Lin
Nature Sensors, **1**, 443–456 (2026)

5. Memristive cellular neural networks for fast in-pixel computing
V. Ravichandran, **Y. Huang**, B. Flannery, T. Molom-Ochir, T. Maurer, S. Asapu, A. Abdel-Maksoud, N. Heermance, R. Yoo, J. Tackie, W. Zhao, Y. Ling, A. Guo, J.J. Yang, Q. Xia
Nature Electronics **9**, 189–199 (2026).
6. Event-based neuromorphic sensing system with flexible haptic sensors and a memristive system on a chip
W. Zhao, **Y. Huang**, A. Tewari, A. J. Rad, A. Zhang, N. Ge, J.J. Yang, M. Hu, S. Majumdar, Q. Xia
Nature Sensors **1**, 163–171 (2026).
7. Radiofrequency signal processing with a memristive system-on-a-chip
Y. Huang*, C. He*, Y. Ling, N. Ge, J.J. Yang, M. Hu, L. Katehi, Q. Xia
Nature Electronics **8**, 587–596 (2025). **(Cover article, July 2025)**
8. A parallel read-write circuit with fast amplitude-adaptive matching scheme to memristor crossbar array
N. Jiang, M. Jiang, H. Huang, L. Yang, **Y. Huang**, Z. Zeng
IEEE Transactions on Circuits and Systems I: Regular Papers **72**, 5898–5911 (2025).
9. Memristor-based hardware accelerators for artificial intelligence
Y. Huang, T. Ando, A. Sebastian, M.F. Chang, J.J. Yang, Q. Xia
Nature Reviews Electrical Engineering **1**, 286–299 (2024).
10. From memristive devices to neuromorphic systems
Y. Huang, F. Kiani, F. Ye, Q. Xia
Applied Physics Letters **122**, 110501 (2023). **(Editor's Pick)**
11. Towards energy-efficient computing hardware based on memristive nanodevices
Y. Huang, V. Ravichandran, W. Zhao, Q. Xia
IEEE Nanotechnology Magazine **17**, 30–38 (2023).
12. MERRC: A memristor-enabled reconfigurable low-power reservoir computing architecture at the edge
F. Nowshin, **Y. Huang**, M.R. Sarkar, Q. Xia, Y. Yi
IEEE Transactions on Circuits and Systems I: Regular Papers **71**, 174–186 (2023).
13. Diffusive memristors with uniform and tunable relaxation time for spike generation in event-based pattern recognition
F. Ye, F. Kiani, **Y. Huang**, Q. Xia
Advanced Materials **35**, 2204778, (2022).
14. Memristor-based HTM spatial pooler with on-device learning for pattern recognition
X. Liu, **Y. Huang**, Z. Zeng, D.C. Wunsch
IEEE Transactions on Systems, Man, and Cybernetics: Systems **52**, 1901–1915 (2020).
15. An associative-memory-based reconfigurable memristive neuromorphic system with synchronous weight training
L. Yang, Z. Zeng, **Y. Huang**
IEEE Transactions on Cognitive and Developmental Systems **12**, 529–540 (2020).
16. Memristor-based circuit design for neuron with homeostatic plasticity
X. Shi, Z. Zeng, L. Yang, **Y. Huang**
IEEE Transactions on Emerging Topics in Computational Intelligence **2**, 359–370 (2018).

17. Memristor-based circuit implementations of recognition network and recall network with forgetting stages
L. Yang, Z. Zeng, **Y. Huang**, S. Wen
IEEE Transactions on Cognitive and Developmental Systems **10**, 1133–1142 (2018).

Conference Publications

1. Hardware-algorithm co-design for hyperdimensional computing based on memristive system-on-chip
Y. Huang, A.J. Rad, Q. Xia
NeurIPS 2024 Workshop Machine Learning with New Compute Paradigms, 1–8 (2024).
2. Three-dimensional memristor-based crossbar architecture for capsule network implementation.
Y. Huang, R. Hu, Z. Zeng
2018 Eighth International Conference on Information Science and Technology, 170–175 (IEEE, 2018).

Invited Talks

1. Hardware-software co-design of memristive systems for edge intelligence
Computer Science Colloquium, City university of Hong Kong, China, 2025
2. Multi-channel testing systems for efficient prototyping of emerging memory devices and arrays
EIPBN 2024 Workshop on Emerging Memory Infrastructure, La Jolla, CA, USA, 2024.

Teaching and Advising

ECE 455/555 Embedded Systems	Fall 2026
University of Tennessee, Knoxville	
Mentor for REU students	Summer 2022/2023
University of Massachusetts Amherst	Project: FPGA programming
Mentor for Senior Design Project	Fall 2018
Missouri University of Science and Technology (Remote)	Project: ML with non-volatile memory

Professional Services

Journal Reviewer for:

- *Science Advances*
- *Nature Communications*
- *Advanced Functional Materials*
- *IEEE Transactions on Cybernetics*
- *IEEE Transactions on Systems, Man and Cybernetics: Systems*
- *IEEE Transactions on Emerging Topics in Computational Intelligence*
- *IEEE Transactions on Circuits and Systems II: Express Briefs*
- *Knowledge-Based Systems*
- *Information Sciences*
- *Neural Networks*
- *Cognitive Computation*
- *Scientific Reports*

Review Committee Member:

- *2026 IEEE International Symposium on Circuits and Systems*